

# EE577B Phase 1 Group 14 Report

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## Cardinal Mesh Router:

## Reset signal operation



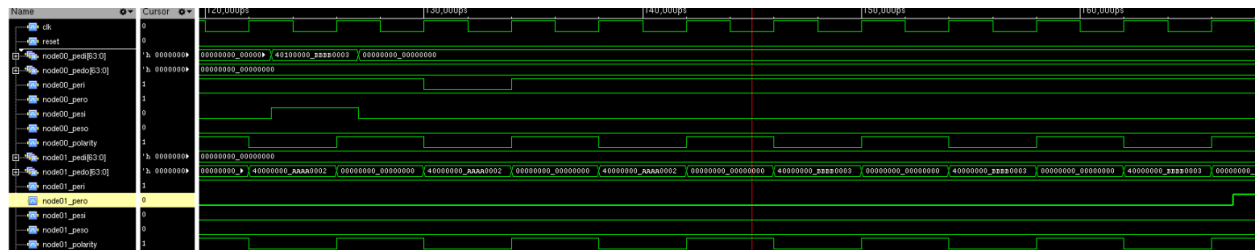
Reset is high from time 0 to 14ns. During this time polarity stays at 0 and doesn't toggle, all the ri signals are 1 meaning the input buffers are empty and ready, and all so signals are 0 meaning nothing is being sent. Once reset goes low at 14ns, polarity starts toggling every 4ns switching between 0 and 1 each clock cycle. This shows the router is initializing everything correctly when reset is applied.

## Handshaking WITHOUT Blocking



A packet with payload AAAA\_0002 is injected at node00's PE input at 43ns when polarity is 0 (even cycle). The header has dirX=1 (East) and hopX=0001 meaning 1 hop left. The packet gets latched into node00's PE input buffer and then sent to the East output. Since node01's west input buffer was empty, the ready signal stayed high the whole time so there was no blocking. The packet then shows up at node01\_pedo at 66ns with hopX decremented to 0000, which means it was delivered correctly. The whole thing took about 23ns which is around 6 clock cycles.

## Handshaking WITH Blocking



At 106ns, node01\_pse is set to 0 which blocks node01 from accepting any packets. At 123ns a packet with payload BBBB\_0003 is injected at node00's PE input. The packet travels through the router but gets stuck because node01 is blocking. At 146ns the packet shows up at node01\_pse but it can't be sent out since pse is still 0. The packet just sits there waiting until 167ns when pse goes back to 1 and the packet is finally released. This shows the blocking behavior where a packet gets held up in the output buffer when the receiving side is not ready.

## Additional 7 test cases

Test 1 — node00 to node03 (3 hops East):

A packet with payload AAAA\_0001 is injected at node00 and routed 3 hops East to node03. The packet arrives at node03 with hopX correctly decremented to 0, confirming eastward XY routing works correctly.

Test 2 — node03 to node00 (3 hops West):

A packet with payload BBBB\_0002 is injected at node03 and routed 3 hops West back to node00. The packet arrives correctly confirming westward routing works.

Test 3 — node00 to node30 (3 hops South):

A packet with payload CCCC\_0003 is injected at node00 and routed 3 hops South to node30. The packet arrives correctly confirming southward Y dimension routing works.

Test 4 — node30 to node00 (3 hops North):

A packet with payload DDDD\_0004 is injected at node30 and routed 3 hops North back to node00. The packet arrives correctly confirming northward routing works.

Test 5 — node00 to node33 (3 hops East + 3 hops South):

A packet with payload EEEE\_0005 is injected at node00 and routed 3 hops East then 3 hops South to node33. This tests XY dimension order routing where the packet first completes X traversal before switching to Y direction.

Test 6 — node33 to node00 (3 hops West + 3 hops North):

A packet with payload FFFF\_0006 is injected at node33 and routed 3 hops West then 3 hops North back to node00. This confirms the full diagonal path works in both directions.

Test 7 — Contention at node00 (node01 and node10 both sending to node00):  
node01 sends a packet West to node00 and node10 sends a packet North to node00 simultaneously. The round-robin arbiter at node00's PE output controller handles the contention by granting node10's packet first (South input has higher priority than West input after reset), then delivering node01's packet on the next available cycle. Both packets arrive successfully demonstrating correct arbiter behavior.

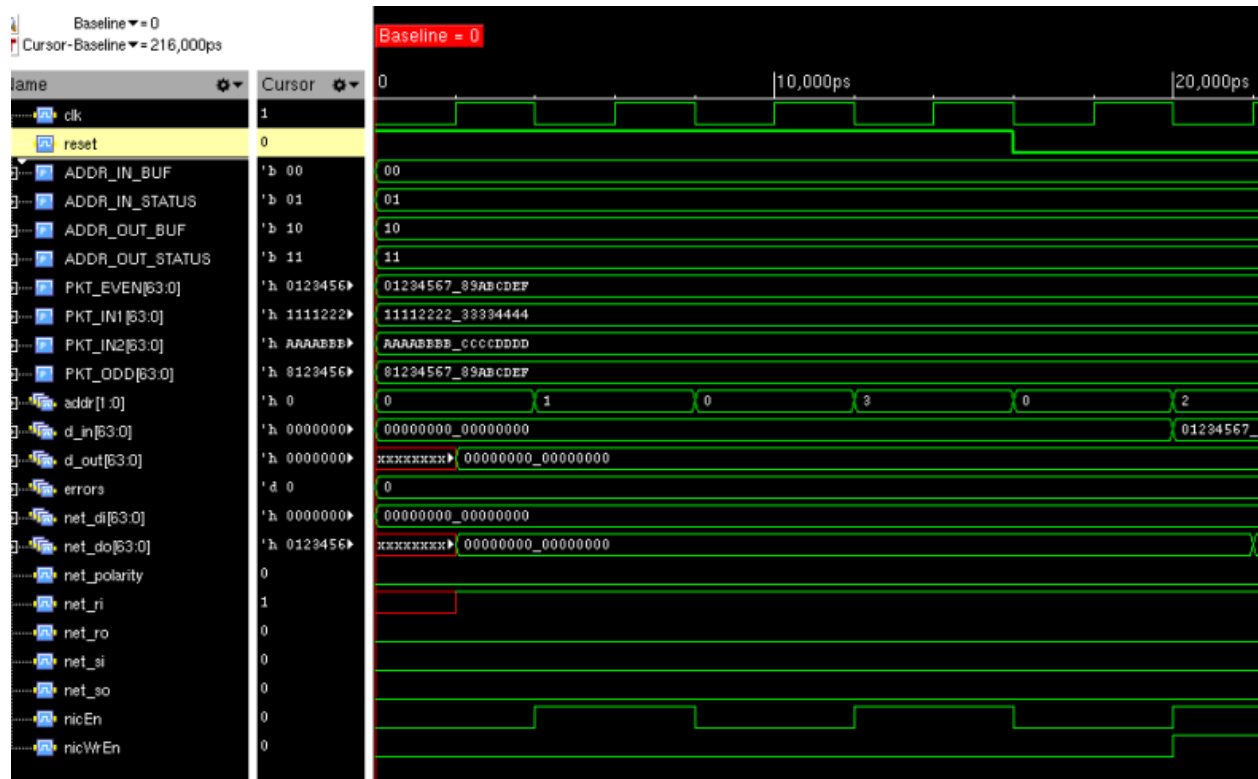
## Correct hop count update

```
assign data_out = (hopX[0] == 1) ?  
    {current_buffer[63:56], current_buffer[55:52] >> 1, current_buffer[51:48], current_buffer[47:0]}  
    :  
    {current_buffer[63:56], current_buffer[55:52], current_buffer[51:48] >> 1, current_buffer[47:0]};
```

The hop value is updated every time a packet moves from an input buffer to an output buffer. Since hop values are unary encoded, decrementing is done by a simple right shift. If hopX[0]==1 the packet is still traveling in the X dimension so only hopX is right shifted. If hopX[0]==0 the packet has finished X traversal so only hopY is right shifted. This can also be seen in the waveforms — for example in Test 1, the packet starts with hopX=0111 at node00 and arrives at node03 with hopX=0000 after 3 right shifts.

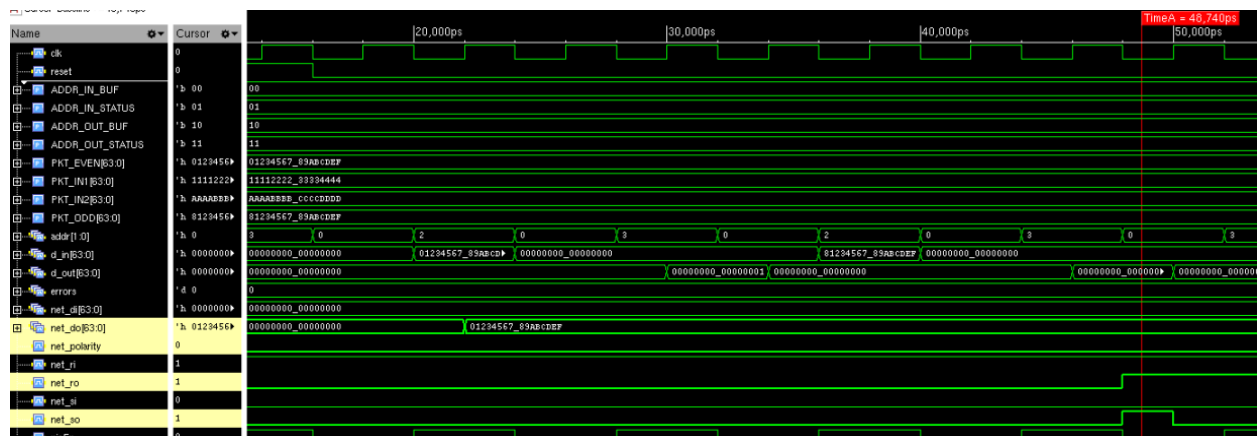
# Cardinal NIC

## Reset Signal Operation



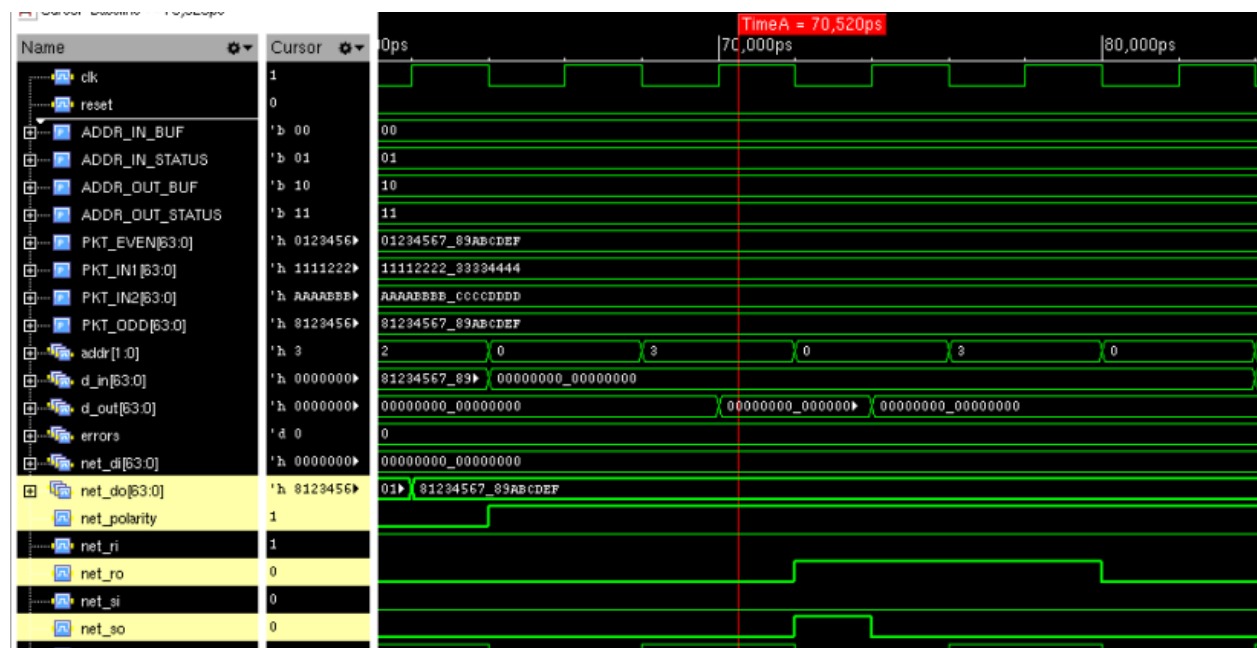
Reset is high from 0 to 16ns. During this time net\_so is 0 meaning the NIC is not sending anything, and net\_ri is 1 meaning the input buffer is empty and ready to receive. Both the input and output status registers are also 0 showing the buffers are empty. Once reset goes low at 16ns the NIC is ready to operate normally. This confirms the NIC initializes correctly when reset is applied.

## Handshaking WITHOUT Blocking



At this point in the waveform, net\_ro=1 indicating the router is ready, net\_polarity=0 which matches the packet's vc bit (bit 63=0 for PKT\_EVEN), and net\_so=1 meaning the NIC is actively sending. The packet 0123\_4567\_89AB\_CDEF is visible on net\_do. Since all handshaking conditions are satisfied at the same time there is no blocking and the packet is transferred to the router in one cycle.

## Handshaking WITH Blocking



At this point in the waveform, PKT\_ODD (8123\_4567\_89ABCDEF) is on net\_do and net\_polarity=1 matches the packet's vc bit. However net\_ro=0 meaning the router is not ready, so net\_so stays 0 and the packet is held in the output buffer. Once net\_ro goes high the NIC immediately asserts net\_so and transmits the packet in one cycle before dropping back to 0. This shows the blocking behavior where the NIC waits for the router to be ready before sending.